

TravelWise AI

Designing an AI-powered multi-modal travel aggregator that helps Indonesian students stop juggling 3+ apps to book a single trip home.

ROLE

Product Manager (Solo)

CONTEXT

Advanced PM Mission · Harisenin.com

TIMELINE

Feb – Apr 2026

SCOPE

Research · Strategy · PRD · Wireframes

Context & Problem

Booking a trip home should take minutes. For students studying away from their hometown ("mahasiswa rantau"), it takes hours — and still ends in regret.

Why I picked this problem

I'm a student in Malang. Every break, I watched classmates — and myself — bounce between Traveloka, Tiket.com, and the KAI app just to plan one trip home, then second-guess the purchase after spotting a cheaper price elsewhere. "I've felt this" isn't a product brief, so I ran a survey to find out whether the friction was real and widespread before designing anything.

What the research showed (n = 30)

96.7%

don't buy from the first app they open

60%

open 3 or more apps per search session

100%

have experienced buyer's remorse after paying

~2M

est. annual transactions from the student segment alone

Problem Statement

Indonesian students who travel between cities — especially those needing multi-modal connections (train → flight) — waste hours across 3+ separate apps, frequently overpay, and have no single platform that handles cross-modal search and booking end-to-end.

Why existing solutions fall short

Platform	Multi-OTA price	Cross-modal routing	Local (KAI) integration	End-to-end booking
Traveloka / Tiket.com	Flights only	No	No	Partial
Skyscanner	Yes	No	No	No (redirect)
Google Flights	Flights only	No	No	No (redirect)
TravelWise AI	Yes	Yes	Yes	Yes

Market sizing (conservative): 6.8M active university students in Indonesia (BPS 2023) × ~30% studying away from home × 2 trips/year ≈ 4M potential transactions/year — before counting migrant workers.

User & Approach

The survey gave me the *shape* of the problem. To keep design decisions anchored to a real human, I distilled the respondent data into a primary persona.



Raka

21 · Engineering student,
Universitas Brawijaya.
From Banjarmasin,
studying in Malang.

*"I always feel like there's a
cheaper ticket somewhere —
I just don't have time to
check five apps."*

GOALS

Get home 2–3× a year on the cheapest reliable route; avoid overpaying; spend less time planning.

FRUSTRATIONS

No flight goes directly to his hometown — he must stitch a train-to-airport leg himself. Hidden fees appear only at checkout.

BEHAVIOR

Opens 3–4 apps per search, screenshots prices to compare, often delays booking and loses the cheap fare.

TECH

Android-first, price-sensitive, comfortable with digital payments and OTAs.

Current journey — where it breaks

1. Trigger Break announced, needs to plan trip home <i>"Here we go again"</i>	2. Search Opens 3–4 OTA apps, compares manually <i>Overwhelmed, tab fatigue</i>	3. Route Manually figures out train → flight transit <i>Confused, anxious about timing</i>	4. Checkout Hidden fees appear; pays anyway <i>Feels tricked</i>	5. After Spots cheaper price elsewhere <i>Regret (100%)</i>
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How I worked — including AI in my workflow

Most case studies describe *what* was decided. I want to show *how* — including an honest account of where I used AI and where my own judgment drove the call.

⚡ AI-Assisted Research Synthesis

After collecting 30 responses, I fed the anonymized open-ended answers into Claude to cluster them into recurring pain-point themes — compressing what would have been hours of manual affinity mapping into roughly 45 minutes. The output gave me four clusters: **price fragmentation, multi-modal complexity, hidden-fee distrust, and buyer's remorse**. The interpretation, prioritization, and every product decision after that were mine. AI accelerated the synthesis step; it did not replace the thinking.

Honest scope note: as a bootcamp mission, prioritization was originally driven by survey data + judgment. The formal RICE model on the next page was built afterward, while developing this case study — and it surfaced a genuinely useful insight (see Reflection).

Prioritization & Scope

With four validated pain points, I needed a defensible way to decide what makes the MVP. I scored every candidate feature with **RICE** (Reach × Impact × Confidence ÷ Effort).

Feature	Reach	Impact	Conf.	Effort	RICE	Decision
Universal Price Aggregator	9	9	90%	5	145	MVP
Price Drop Alert	7	8	80%	4	112	MVP
Smart Cross-Modal Route Finder	7	10	70%	9	54	MVP
AI Itinerary Builder	5	7	60%	8	26	Phase 2
Student Promo Verification	5	6	50%	9	17	Phase 2

The judgment call RICE couldn't make for me

By raw score, the Cross-Modal Route Finder (54) ranks *below* Price Drop Alert (112) because its effort is high. But I still made it the core MVP feature — because it's the one capability no competitor offers, and it's the actual reason a user would choose us over Traveloka. **RICE is a decision aid, not a decision maker.** Shipping only the high-score, low-effort features would have produced a faster product with no reason to exist.

The resulting MVP — 3 features

Feature	Priority	Why it's in
Universal Price Aggregator	High	Solves the core fragmentation pain; shows final (post-tax) prices in one screen.
Smart Cross-Modal Route Finder	High	The differentiator — auto-matches train+flight with safe transit buffers, one checkout.
Price Drop Alert	Medium	Directly attacks buyer's remorse; builds an engagement loop that brings users back.

Explicitly out of scope (and why)

- **AI Itinerary Builder & Student Promo** — high effort, lower fit for the priority segment; deferred to Phase 2.
- **Hotel booking & international routes** — focus stays on domestic transport tickets until traction is proven.
- **Web app** — mobile-first only, matching how the segment actually searches and books.

Defining what *not* to build was as deliberate as the MVP itself — each cut has a reason I can defend.

Requirements & AI Feature Spec

The PRD defines 5 user stories with acceptance criteria. Below is the requirement I'm proudest of — because it shows how technical understanding makes a PM's requirements buildable.

User stories at a glance

ID	Story	Priority
US-01	Search & compare ticket prices across OTAs in one screen	Critical
US-02	Book a cross-modal route (train + flight) in one checkout	Critical
US-03	Set a price-drop alert on a watched route	High
US-04	See true total price up front — no hidden fees	Critical
US-05	Register & onboard in 3 steps or fewer	Medium

⚡ Spotlight — Writing requirements for the AI matching engine (US-02)

A generic PM writes: *"AI finds the cheapest route."* Because I've worked hands-on with ML pipelines, I could specify requirements an engineer can actually build to:

- Enforce a **minimum 90-minute transit buffer** between modes — accounting for train delays and airport transfer, not just theoretical connection time.
- Rank options by a **composite score** (price 60% · total time 30% · transit risk 10%) rather than cheapest-first, because the cheapest route is often the riskiest.
- **Degrade gracefully**: if one OTA's data is unavailable, show partial results with a clear notice instead of failing the whole search.
- **Cache prices with a 5–15 min TTL** — fresh enough to stay accurate at checkout, long enough to avoid hammering OTA APIs on every keystroke.

Knowing *why* each number exists is what separates a requirement that ships from one that bounces back from engineering.

Success metrics

Metric	Target	How measured
Search-to-booking conversion (North Star)	≥ 15%	Funnel analytics
Users not opening another OTA after use	≥ 80%	In-app exit survey
Cross-modal bookings as % of total	≥ 30%	Transaction logs
Month-2 retention	≥ 30%	Cohort analysis

North Star rationale: search-to-booking conversion validates both that aggregation surfaced a relevant result *and* that price transparency earned enough trust to complete the purchase — it captures the whole value proposition in one number.

Outcome & Reflection

Honest framing

This was a research-and-design deliverable, not a launched product. The points below are validated learning and a concrete validation plan — not live performance metrics. I'd rather show honest reasoning than invent numbers.

What the research validated

A 96.7% problem-confirmation rate across 30 respondents — for a pain point this specific — is a strong signal to continue to solution design. The fragmentation and multi-modal pains are real and currently unserved by any major player in Indonesia.

What I'd test before building anything

Before writing a line of matching-engine code, I'd run a **concierge MVP**: manually handle 10–20 real cross-modal searches for users and watch where the friction lives. Hypothesis to test — if users repeatedly ask "is this connection safe?", the UI needs an explicit "why this transit is safe" layer before the algorithm matters.

What I'd do differently

1 • Validate API access before scoping

The whole MVP assumes Traveloka, Tiket.com, and KAI expose usable APIs. In reality, Indonesian OTAs are often restrictive. I should have run a short technical-discovery sprint — even a few messages to developers at these companies — before locking "universal aggregator" as the core. This is the single biggest risk to the product, and I scoped around it instead of into it.

2 • Add qualitative depth to the research

30 survey responses told me *what* happens but not *why*. Why do 60% open 3+ apps — distrust, habit, or loyalty points? Even 5 short interviews alongside the survey would have sharpened both the persona and the solution.

3 • Applying RICE later changed how I see prioritization

Scoring formally *after* the fact revealed that my gut-chosen core feature was actually the lowest-RICE of the three MVP items. Rather than invalidating the choice, it taught me to **separate "highest score" from "most strategic"** — and to state that trade-off explicitly instead of hiding behind a framework.

What this taught me about being a PM

Out of scope is a product decision, not a limitation. The strongest part of this project wasn't the features I included — it was being able to defend every feature I cut. And using AI well, I learned, means using it to move faster on the mechanical work so I can spend more judgment on the decisions only a human should make.